

Asynchronous Multimodal View Synthesis with Probabilistic Temporal Encoding

ZIHAO ZOU, University of North Carolina, Chapel Hill, USA

JIAMING LIU, Stanford University, USA

HENGYU LIAN, University of North Carolina, Chapel Hill, USA

NAGASAI THADISHETTY, University of North Carolina, Chapel Hill, USA

PRANEETH CHAKRAVARTHULA, University of North Carolina, Chapel Hill, USA

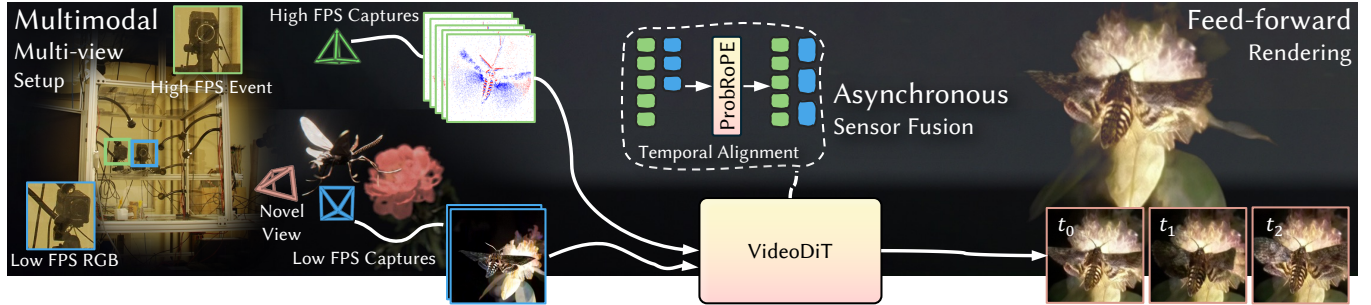


Fig. 1. *Feed-forward Asynchronous Multimodal Sensor Fusion.* We introduce a feed-forward video diffusion transformer, D-FUSE, that performs asynchronous, multimodal view synthesis of dynamic scenes. The core of our approach is a probabilistic temporal encoding ProbRoPE that addresses the temporal misalignment between asynchronous sensor streams to generate temporally coherent high-speed videos. Here, we demonstrate a feed-forward high-speed reconstruction of a hummingbird hawk moth from fusing irregular captures from a low-frame-rate RGB camera and a high-frame-rate event camera.

Multiview capture and rendering of dynamic scenes and fast phenomena typically rely on tightly synchronized high-speed camera arrays, making such systems expensive and difficult to scale. Multimodal sensing offers a practical low-cost alternative by combining complementary signals, such as RGB cameras for appearance and event cameras for temporal resolution. However, these sensors are often spatially separated and operate at different sampling rates, producing asynchronous and heterogeneous observations that are difficult to align across viewpoints. Existing approaches either support asynchronous multimodal fusion through costly per-scene optimization, or enable feed-forward generalization but only for synchronized single-modality inputs or static scenes.

In this work, we show that both asynchronous multimodal fusion and feed-forward novel view synthesis can be achieved in a single framework. We present D-FUSE, a feed-forward diffusion transformer for spatiotemporal interpolation and novel view synthesis from spatially separate asynchronous multimodal observations. Central to our approach is ProbRoPE, a probabilistic extension of rotary position embedding (RoPE) that represents each observation as a probability distribution over its capture interval, enabling

alignment across viewpoints and sampling rates. Our method operates directly on irregular asynchronous inputs and produces temporally consistent novel views under fast motion and sparse viewpoints, significantly outperforming existing approaches.

CCS Concepts: • **Computing methodologies** → **Image-based rendering; Reconstruction.**

Additional Key Words and Phrases: novel view synthesis, multimodal sensor fusion, event cameras, video diffusion, neural rendering

ACM Reference Format:

Zihao Zou, Jiaming Liu, Hengyu Lian, Nagasai Thadishetty, and Praneeth Chakravarthula. 2026. Asynchronous Multimodal View Synthesis with Probabilistic Temporal Encoding. In *SIGGRAPH Asia 2026 Conference Papers (SA Conference Papers '26)*, December 01–04, 2026, Kuala Lumpur, Malaysia. ACM, New York, NY, USA, 13 pages. <https://doi.org/10.1145/3829340.3842258>

1 Introduction

Capturing and rendering high-speed dynamic scenes from multiple viewpoints is a fundamental task across many domains, including telepresence in remote communication, virtual production in film, and behavioral analysis in biological imaging. Conventional solutions rely on an array of high-speed cameras, but the recorded data grows as a product of frame rate, viewpoint count, and duration, limiting recordings to only a few seconds and making such arrays difficult to scale. A more practical choice is multimodal sensing, where heterogeneous sensors with complementary abilities jointly capture a dynamic scene. For example, RGB cameras capture rich color and appearance, and event cameras [Gallego et al. 2020] deliver microsecond temporal resolution with low bandwidth requirements. Combining them provides rich measurements in both space and

Authors' Contact Information: Zihao Zou, zihaozou@cs.unc.edu, University of North Carolina, Chapel Hill, Chapel Hill, NC, USA; Jiaming Liu, jiamliu@stanford.edu, Stanford University, Stanford, CA, USA; Hengyu Lian, lianhy0@cs.unc.edu, University of North Carolina, Chapel Hill, Chapel Hill, NC, USA; Nagasai Thadishetty, nthadishetty1@gsu.edu, University of North Carolina, Chapel Hill, Chapel Hill, NC, USA; Praneeth Chakravarthula, cpk@cs.unc.edu, University of North Carolina, Chapel Hill, Chapel Hill, NC, USA.

SA Conference Papers '26, Kuala Lumpur, Malaysia

© 2026 Copyright held by the owner/author(s).

This is the author's version of the work. It is posted here for your personal use. Not for redistribution. The definitive Version of Record was published in *SIGGRAPH Asia 2026 Conference Papers (SA Conference Papers '26)*, December 01–04, 2026, Kuala Lumpur, Malaysia, <https://doi.org/10.1145/3829340.3842258>.

time. However, when deployed together, these sensors are typically spatially separated and operate at different frame rates: RGB cameras capture at tens of frames per second, while event cameras produce streams of events at microsecond resolution. Faithfully aligning such asynchronous and heterogeneous measurements across space and time remains a key challenge.

A common approach to this problem is *per-scene optimization*. Methods [Feng et al. 2025; Ma et al. 2023; Rudnev et al. 2025; Zou et al. 2025] based on neural radiance fields [Mildenhall et al. 2021] or 3D Gaussian splatting [Kerbl et al. 2023] fuse asynchronous multimodal data by iteratively optimizing a scene representation separately for each scene. Although they achieve impressive visual quality, these methods require lengthy per-scene training, depend on dense, carefully aligned multi-view input, and are unable to generalize to unseen regions. As a result, they cannot generalize across scenes and are unable to handle sparse views.

An alternative line of work avoids per-scene optimization through *feed-forward novel view synthesis (NVS)*, where a single model is trained on diverse data to generalize across scenes. By leveraging learned priors over scene appearance and geometry, these methods can directly predict the scene representation [Lin et al. 2026] or novel views [Van Hoorick et al. 2024; Yu et al. 2025]. However, they are typically restricted to either a single modality (e.g., RGB) or static scenes. Overall, no existing approach addresses feed-forward NVS of dynamic scenes from asynchronous multimodal multi-view sensors.

In this work, we fill this gap with a video diffusion transformer (VideoDiT) [Peebles and Xie 2023]. Extending VideoDiT to our setting faces challenges in *temporal aliasing in positional encoding*. VideoDiT compresses each input video stream into spatiotemporal patches, and assigns each patch a positional code [Su et al. 2024] to represent its index in the frame sequence. This approach, however, ignores the physical frame rate and the temporal extent of each patch, leading to temporal aliasing when streams with different frame rates are fused together. This aliasing results in ghosting and unrealistic motion in the synthesized views. To address this issue, we introduce *ProbRoPE*, which represents the temporal position as a distribution over time rather than a deterministic point. It faithfully encodes the physical frame rate and the temporal extent of each patch, allowing accurate fusion across frame rates. With this encoding, we present D-FUSE, which faithfully fuses asynchronous multimodal multi-view data for novel view synthesis in a fully feed-forward manner.

We conduct extensive experiments to show that ProbRoPE effectively addresses the temporal aliasing problem, enabling faithful motion and geometry reconstruction. We demonstrate that D-FUSE beats both existing per-scene methods and feed-forward approaches in terms of visual and perceptual quality on synthetic data. We further validate D-FUSE on real-world data captured by a multimodal system consisting of an RGB and an event camera, showing promising potential for various applications such as biological imaging.

Project materials and code are available here.

2 Background and Related Work

2.1 Novel View Synthesis of Dynamic Scenes

Conventional approaches for dynamic NVS typically rely on dense, tightly synchronized high-speed camera arrays [Broxton et al. 2020; Joo et al. 2015; Lombardi et al. 2019; Wilburn et al. 2005]. However, these systems are expensive and difficult to deploy. Multimodal sensor fusion, e.g. combining standard RGB with high-speed event cameras [Gallego et al. 2020], has emerged as a promising alternative. Existing multimodal methods either rely on lengthy *per-scene optimization* (failing to generalize across scenes), or are *feed-forward methods* that remain confined to single-modality or strictly synchronized inputs.

2.2 Per-scene Optimization for Dynamic NVS

Per-scene optimization fits a scene-specific representation, primarily Neural Radiance Fields (NeRF) [Mildenhall et al. 2021] or 3D Gaussian Splatting (3DGS) [Kerbl et al. 2023], from scratch. Barron et al. [2021] and Yu et al. [2024] extended these to include anti-aliasing techniques. Later works further extended these methods to dynamic scenarios via time-conditioned implicit fields [Du et al. 2021; Gao et al. 2021; Li et al. 2022, 2021; Park et al. 2021a,b; Pumarola et al. 2021; Xian et al. 2021] or explicit spatiotemporal decompositions [Cao and Johnson 2023; Fang et al. 2022; Fridovich-Keil et al. 2023; Liu et al. 2022a; Shao et al. 2023; Song et al. 2023]. More recently, dynamic extensions of 3DGS [Bae et al. 2024; Duan et al. 2024; Huang et al. 2024; Li et al. 2024; Lin et al. 2024; Luiten et al. 2024; Wang et al. 2025; Wu et al. 2024; Yang et al. 2024, 2023; Zhang et al. 2025] have further evolved these techniques, enabling real-time rendering with competitive reconstruction performance.

Based on these dynamic NeRF/3DGS frameworks, a line of research has explored integrating additional sensors to provide stronger spatial or temporal supervision. Attal et al. [2021]; Qingming et al. [2025] leverage depth sensors to provide direct geometric supervision. Malik et al. [2023]; Mu et al. [2024] utilize Single-Photon Avalanche Diodes (SPADs) to capture time-resolved transient histograms, enabling 3D reconstruction even under extreme sparsity. Cannici and Scaramuzza [2024]; Klenk et al. [2023]; Rudnev et al. [2025]; Zou et al. [2025] integrate event cameras to capture high-speed dynamics without motion blur.

Despite achieving impressive visual quality, these per-scene optimization methods require exhaustive optimization, often taking hours to days [Rudnev et al. 2025; Zou et al. 2025] for every new scene, and fail to generalize to unseen regions, making them impractical for real-world applications.

2.3 Feed-forward Methods for Generalizable NVS

An alternative paradigm explores feed-forward and generalizable architectures. By leveraging strong geometry and appearance priors learned from large-scale datasets, these models can directly infer 3D representations and synthesize novel views from sparse inputs in a feed-forward manner, entirely bypassing lengthy optimization.

Regression-based works achieve this by directly outputting the novel views [Jin et al. 2024] or intermediate representations [Charatan et al. 2024; Chen et al. 2024; Jiang et al. 2025; Szymanowicz et al. 2024; Wang et al. 2021; Yu et al. 2021] from input images. Recent

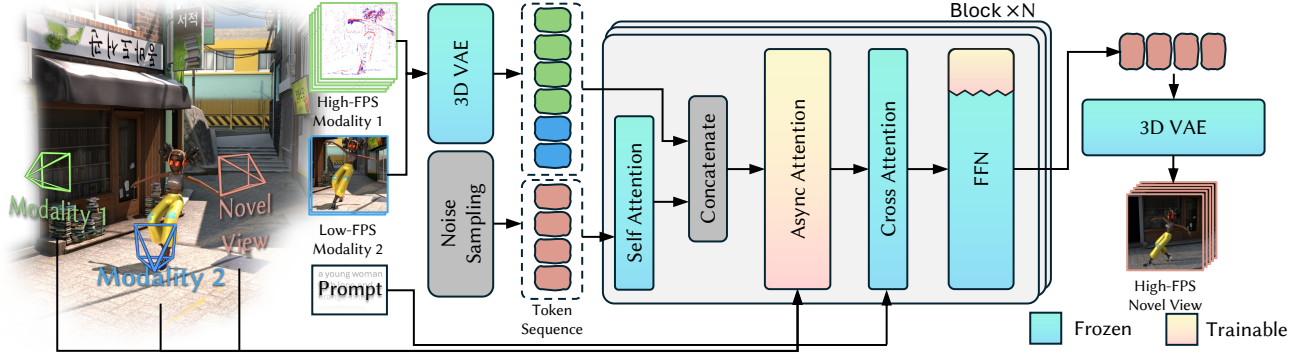


Fig. 2. *Overview of D-FUSE.* D-FUSE is a video diffusion model for asynchronous, multimodal view synthesis of dynamic scenes. It takes as input a set of asynchronous, multi-sensor video streams (e.g., RGB and events), and outputs a novel-view video at a user-specified viewpoint. At its core, D-FUSE leverages AsyncAttention, an attention operation that fuses asynchronous, multimodal inputs and conditions the video diffusion process.

works push this paradigm into the dynamic domain by predicting either dynamic radiance fields [Miao et al. 2024; Tian et al. 2023] or 4D Gaussian representations [Liang et al. 2024; Ren et al. 2024]. However, these methods cannot generalize to occluded content, resulting in degraded visual quality.

To overcome this limitation, an expanding paradigm leverages *generative* priors, exploiting pre-trained generative models to synthesize coherent novel views even at unobserved viewpoints. Gao et al. [2024]; Wu et al. [2025b] leverage diffusion models to synthesize novel views of static scenes. Lu et al. [2025] further extend this strategy to fuse depth data. Moreover, Bai et al. [2025]; Park et al. [2025]; Van Hoorick et al. [2024]; Xie et al. [2026]; Yu et al. [2025] finetune a pre-trained video diffusion model for dynamic NVS, and achieve impressive visual quality. However, these methods are almost exclusively tailored for homogeneous, tightly synchronized RGB inputs, and their ability to process fast or complex motion is fundamentally bottlenecked by the low frame rate of standard cameras.

3 Approach

We begin by describing the architecture of video diffusion transformers, sensor fusion, and the standard positional encoding scheme in

Section 3.1, and expose their limitations in handling asynchronous streams. We then introduce D-FUSE in Section 3.2, where asynchronous inputs are fused within the attention layers of a video diffusion transformer. This is possible due to ProbRoPE, our novel temporal encoding that represents each observation as a distribution over its temporal span, enabling asynchronous cross-modal fusion.

3.1 Preliminaries

Video diffusion transformers (VideoDiT). A VideoDiT [Peebles and Xie 2023] consists of a 3D variational autoencoder (3D VAE) [Kingma and Welling 2014] and a transformer-based denoiser that operates in the VAE latent space [Rombach et al. 2022]. As illustrated in Fig. 3 (Top), given an input video $\mathbf{I}$, the 3D VAE encoder compresses it into a latent grid $\mathbf{Z}$. Each latent vector in this grid, called a *token* $\mathbf{x}$, represents pixels over a spatiotemporal patch. As illustrated in Fig. 3 (Bottom), the diffusion process is an ODE over $\tau \in [0, 1]$ that transports Gaussian noise $\mathbf{Z}_1 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ to a data sample $\mathbf{Z}_0$, with the velocity field evaluated by the transformer denoiser $\mathbf{v}_\phi(\mathbf{Z}_\tau, \tau)$ at step τ [Lipman et al. 2022; Liu et al. 2022b]. The core of the transformer is the attention operation. Given a flattened token sequence $\{\mathbf{x}_i\}_{i=1}^N$, we project them into query vectors $\{\mathbf{q}_i\}_{i=1}^N$, key vectors $\{\mathbf{k}_j\}_{j=1}^N$, and value vectors $\{\mathbf{v}_j\}_{j=1}^N$ via learned projection matrices. The output $\mathbf{o}_i$ at token i is then computed as a weighted sum of value vectors, with weights given by the softmax-normalized attention scores s_{ij} between queries and keys:

$$\mathbf{o}_i = \sum_j \text{softmax}_j(s_{ij}) \mathbf{v}_j, \quad s_{ij} = \frac{\mathbf{q}_i^\top \mathbf{k}_j}{\sqrt{d_k}}, \quad (1)$$

where d_k is the key dimension. This operation fuses and exchanges information across all tokens in the sequence, allowing global modeling of the spatiotemporal structure of the input video stream.

Rotary Position Embedding (RoPE). However, since we flatten the tokens into a sequence, the attention operation is blind to the spatiotemporal position of $\mathbf{x}_i$ in the grid $\mathbf{Z}$. RoPE [Su et al. 2024] injects positional information by rotating queries and keys. We divide each query/key vector $\mathbf{q}_i, \mathbf{k}_j \in \mathbb{R}^{d_k}$ into $d_k/2$ two-dimensional subspaces, indexed by $l = 1, \dots, d_k/2$. In the l -th subspace, RoPE rotates the 2D

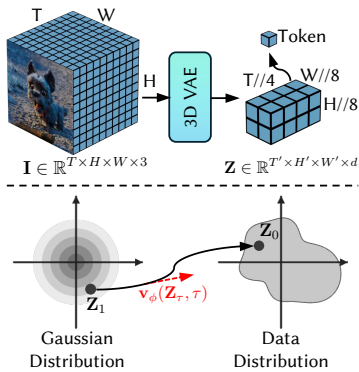


Fig. 3. *Latent Compression (Top) and Diffusion Process (Bottom)*

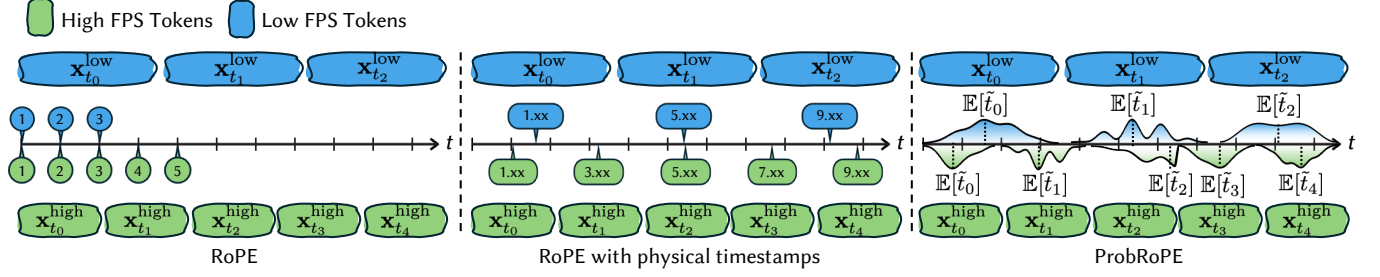


Fig. 4. *Comparison of Different RoPE Embeddings.* We compare how the temporal information is encoded in the original RoPE [Su et al. 2024], RoPE with physical timestamps, and our proposed ProbRoPE. The original RoPE encodes the frame index, ignoring the physical frame rates of multimodal tokens. RoPE with physical timestamps encodes the timestamp, but fails to model the physical time span of each token. In contrast, our ProbRoPE encodes the expected timestamp over the time span of each token, effectively modeling the asynchrony of multimodal inputs.

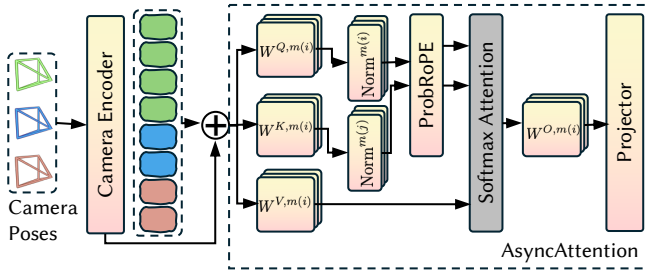


Fig. 5. *AsyncAttention.* The multiview and multimodal information is input to the AsyncAttention as a single token sequence (colored by source camera). The camera poses are encoded by a zero-initialized convolution and are added to the token features before the projections W (stacked blocks denoting per-modality parameters). ProbRoPE replaces the standard RoPE before the softmax attention, aligning streams running at different rates.

query $q_{i,l}$ and key $k_{j,l}$ by position-dependent angles $\theta_l p_i$ and $\theta_l p_j$, where $p_i \in \mathbb{Z}_{\geq 0}$ is the $\mathbf{x}_i$ position in $\mathbf{Z}$ and $\theta_l = 10000^{-2(l-1)/d_k}$ is a fixed frequency. Treating each 2D subspace as a complex number $q_{i,l}, k_{j,l} \in \mathbb{C}$, the rotation can be written in Euler form as

$$\tilde{q}_{i,l} = q_{i,l} e^{i\theta_l p_i}, \quad \tilde{k}_{j,l} = k_{j,l} e^{i\theta_l p_j}, \quad l \in \{1, \dots, d_k/2\}. \quad (2)$$

Substituting the rotated queries and keys into Eq. (1) gives a per-channel contribution $s_{ij,l}$ and the full attention score s_{ij} :

$$s_{ij,l} = \text{Re}[q_{i,l} \tilde{k}_{j,l} e^{i\theta_l(p_i - p_j)}], \quad s_{ij} = \frac{1}{\sqrt{d_k}} \sum_l s_{ij,l}, \quad (3)$$

where $\text{Re}[\cdot]$ is the real part. A natural interpretation of RoPE is that the relative position $p_i - p_j$ is encoded as a spectrum of frequencies $\{\theta_l\}$: higher θ_l respond sensitively to close-by tokens, while lower θ_l respond to distant tokens. Replacing p_i with a spatiotemporal index, e.g., the frame index along the temporal axis, enables RoPE to encode spatiotemporal positions. However, this treats time as a uniform integer grid. When sensors run at different rates, tokens represent different temporal extents, but are assigned frame indices with the same stride.

3.2 Asynchronous Sensor Fusion VideoDiT

As illustrated in Fig. 2, we introduce D-FUSE, a VideoDiT that takes a set of asynchronous multi-view, multimodal streams as conditions and synthesizes a high-speed novel-view RGB video. We build D-FUSE on a pretrained video diffusion model. Besides the original self-attention across denoising tokens and cross-attention between denoising tokens and the text prompt, we add an extra attention module, AsyncAttention, to generate video conditioned on asynchronous multi-view, multimodal streams.

As illustrated in Fig. 5, AsyncAttention modifies the standard multi-head self-attention in three ways. First, following Bai et al. [2025]; Gao et al. [2024], we inject the camera parameters of each token by encoding them with a zero-initialized convolution (*Camera Encoder* in the figure) and adding the result to the token features before the projections. This allows the attention to be aware of the camera poses and align features across views. Second, we instantiate modality-specific parameters [Esser et al. 2024] for the query, key, and value projections. During the forward pass, each token is routed through modality-specific transformations W , illustrated as stacked blocks in Fig. 5. Employing modality-specific parameters avoids suboptimal representation of multimodal sensor streams while still allowing cross-modal fusion via the attention. Third, we replace the standard RoPE with ProbRoPE to faithfully fuse asynchronous multi-view streams running at different frame rates, which we develop in the remainder of this section.

While existing methods can handle synchronous multi-modal multi-view inputs, they do not address the temporal aliasing across asynchronous streams. Sensor modalities operate at very different native rates: an RGB camera captures at tens of frames per second, while an event sensor produces measurements at an effective rate of kilohertz. For instance, consider the example in Fig. 4. A low-rate stream (blue) produces three wide tokens over the same physical time interval that a high-rate stream (green) covers with five narrow ones; each token’s width depicts its frame-rate-dependent span Δt_i .

Conventional positional encoding fails on this setup in two distinct ways. Standard RoPE (Fig. 4 left) assigns each token a position equal to its frame index, ignoring both the time t_i at which the token was captured and its time span Δt_i . As a result, two tokens corresponding to different physical times receive the same positional code. Under Eq. (3), the early high-rate tokens then attend blindly

to the low-rate tokens, while the later high-rate tokens, having no index-matched counterpart, barely participate in cross-stream attention. A natural fix is to replace the index with the physical timestamp t_i , placing all tokens on a shared timeline [HaCohen et al. 2024; Wu et al. 2025a]. This aligns tokens across streams in time. However, as shown in Fig. 4 middle, the second blue token receives the same code as the third green token. Under Eq. (3), they attend strongly to each other and fuse information that lies outside their respective time spans. Both failures are caused by the fact that the time of a token is treated as a point rather than an interval the token spans. Therefore, a faithful temporal encoding should reflect not only when a token was captured, but also how long it covers.

We introduce ProbRoPE (Fig. 4 right), a probabilistic temporal encoding which replaces the deterministic temporal position p in Eq. (3) with a random number $\tilde{t}_i \sim P_i$, where P_i is the probability distribution supported on the token’s time span $[t_i - \Delta t_i/2, t_i + \Delta t_i/2]$. Under this formulation, Eq. (3) also becomes a random variable. Since each token represents its entire span, the appropriate score between two region-valued tokens is the *expectation* of Eq. (3). For each channel l , we define the per-channel score expectation as:

$$\begin{aligned} \mathbb{E}[s_{ij,l}] &\stackrel{\text{def}}{=} \mathbb{E}_{\tilde{t}_i, \tilde{t}_j} \left[\text{Re} \left[q_{i,l} \bar{k}_{j,l} e^{i\theta_l(\tilde{t}_i - \tilde{t}_j)} \right] \right] \\ &= \text{Re} \left[q_{i,l} \bar{k}_{j,l} \cdot \mathbb{E}_{\tilde{t}_i, \tilde{t}_j} \left[e^{i\theta_l \tilde{t}_i} \cdot e^{-i\theta_l \tilde{t}_j} \right] \right] \\ &= \text{Re} \left[q_{i,l} \bar{k}_{j,l} \cdot \mathbb{E}_i \left[e^{i\theta_l \tilde{t}_i} \right] \cdot \overline{\mathbb{E}_j \left[e^{i\theta_l \tilde{t}_j} \right]} \right]. \end{aligned} \quad (4)$$

We reparameterize each expectation around the token midpoint by setting $u = \tilde{t} - t$, which yields

$$\begin{aligned} \mathbb{E}[e^{i\theta_l \tilde{t}}] &= \int_{-\Delta t/2}^{\Delta t/2} e^{i\theta_l(t+u)} P(t+u) du \\ &= \underbrace{e^{i\theta_l t}}_{\text{standard RoPE}} \cdot \underbrace{\int_{-\Delta t/2}^{\Delta t/2} e^{i\theta_l u} P(t+u) du}_{\lambda_l : \text{complex modulator}}. \end{aligned} \quad (5)$$

The expectation factors into the standard RoPE $e^{i\theta_l t}$, scaled by a modulator λ_l . By Fourier scaling [Oppenheim et al. 1997], λ_l decays with higher θ_l that exceed the token’s temporal sampling frequency $1/\Delta t$, acting as an **anti-aliasing** filter. While anti-aliasing filters have been explored for 3D rendering [Barron et al. 2021; Yu et al. 2024], this is the first work to extend it to attention mechanism, enabling robust alignment of asynchronous sensor streams. More details can be found in the supplementary material.

After substituting Eq. (5) into Eq. (4) and writing the modulator in polar form as $\lambda_{i,l} = |\lambda_{i,l}| e^{i\theta_l \psi_{i,l}}$, where $|\lambda_{i,l}|$ is its magnitude and $\psi_{i,l}$ is its phase, the per-channel score becomes

$$\begin{aligned} \mathbb{E}[s_{ij,l}] &= \text{Re} \left[q_{i,l} \bar{k}_{j,l} \cdot e^{i\theta_l t_i} \lambda_{i,l} \cdot \overline{e^{i\theta_l t_j} \lambda_{j,l}} \right] \\ &= \text{Re} \left[q_{i,l} \bar{k}_{j,l} \cdot |\lambda_{i,l}| |\lambda_{j,l}| \cdot e^{i\theta_l(t_i - t_j + \psi_{i,l} - \psi_{j,l})} \right] \\ &= |\lambda_{i,l}| |\lambda_{j,l}| |q_{i,l} \bar{k}_{j,l}| \cos((t_i - t_j + \psi_{i,l} - \psi_{j,l})\theta_l). \end{aligned} \quad (6)$$

The magnitudes $|\lambda_{i,l}|$ and $|\lambda_{j,l}|$ act as per-channel attention amplifiers, while the phases $\psi_{i,l}$ and $\psi_{j,l}$ shift the effective relative position between tokens. Therefore, the score calculation no longer blindly

relies on the token position, but considers the token’s temporal span.

Different choices of P yield different modulators. For example,

- **Uniform** $\rightarrow \lambda_l = \text{sinc}(\theta_l \Delta t/2) \in \mathbb{R}$, where the score is scaled by a sinc function that depends on Δt .
- **Gaussian** $\rightarrow \lambda_l = e^{-\sigma_l^2 \theta_l^2/2} \in \mathbb{R}$, with σ_l predicted from Δt_l by an MLP. The score decays as a Gaussian in θ_l , giving a smooth low-pass over the RoPE frequencies.
- **Learned** $\rightarrow$ **complex modulator**. We let λ_l be learned by a small MLP conditioned on Δt and $\mathbf{x}$. This design allows the distribution to be shaped flexibly by the data. Before applying the modulation, we rotate the query and key to a learned orthogonal basis [Fei et al. 2022; Li et al. 2025a]. We empirically find that this improves the performance of the learned modulator.

We ablate these design choices in Section 6, and provide details on the derivation, implementation, and analysis of ProbRoPE in the supplementary material.

We address the temporal aliasing of asynchronous videos in the transformer, thus enabling asynchronous multimodal view synthesis with VideoDiT.

4 Implementation

For the modality, we select **RGB images** and **event streams**. RGB provides dense spatial appearance but limited temporal resolution, while event streams provide high temporal resolution at low bandwidth but lack direct color information. This combination challenges the model in cross-modal fusion and temporal asynchrony. For the modulator, we use the **Learned modulator**, which we find to perform best in Section 6.

Training Data. Due to the lack of paired multi-view multimodal video datasets, we assemble a large-scale collection of multi-view RGB video datasets, including Bai et al. [2024], Shapovalov et al. [2023], and Van Hoorick et al. [2024]. We synthesize the events from the RGB videos using the V2E simulator [Hu et al. 2021]. We also include a large-scale static multi-view dataset Reizenstein et al. [2021], treating it as single-modal single-frame video, to enrich the generalization to real-world scenarios. In total, we have 150,640 sequences and 18,693,786 frames across 42,379 distinct scenes. Each sequence is captioned with Qwen3.5-8B [Qwen Team 2026] to provide text conditioning.

Training Details. We implement D-FUSE with a pretrained Wan2.1-T2V-1.3B [Wan et al. 2025] backbone and fine-tune the full AsyncAttention block and the event’s modality-specific FFN while keeping the others frozen. We initialize the weights to be identical to the self-attention block. Training uses the standard rectified flow objective [Liu et al. 2022b], a batch size of 32, AdamW [Loshchilov and Hutter 2017] with learning rate $1e-4$. We train for 50k iterations on 8x A100 GPUs over approximately 21 days. During training, the target video is sampled at the full event rate, while the conditioning RGB stream is randomly subsampled to a frame rate in $\{1/2, 1/4, 1/8\}$ of that rate, exposing the model to varying degrees of temporal asynchrony. Additional implementation details can be found in the supplementary material.

Table 1. **Quantitative evaluation on synthetic Kubric4D [Van Hoorick et al. 2024]**. We condition on RGB and event streams captured from distinct viewpoints of the same scene. The event stream is kept at its native frame rate; RGB streams are temporally subsampled to 1/2, 1/4, and 1/8 of that rate, probing performance under increasing temporal asynchrony between the two modalities. The output video is rendered at the full event rate, corresponding to 2×, 4×, and 8× RGB upsampling. Per-scene optimization baselines are fit to each scene and, due to their training cost, are evaluated on 16 scenes (Kubric4D-16) with 3 RGB views and 1 event view. Feed-forward diffusion baselines are evaluated on the full 100-scene test set (Kubric4D-100) with 1 RGB view and 1 event view; they receive an RGB stream first upsampled to the event rate via FILM [Reda et al. 2022].

		2× Upsampling					4× Upsampling					8× Upsampling				
Method		PSNR↑	LPIPS↓	FID↓	FVD↓	MUSIQ↑	PSNR↑	LPIPS↓	FID↓	FVD↓	MUSIQ↑	PSNR↑	LPIPS↓	FID↓	FVD↓	MUSIQ↑
Kubric4D-16 3 RGB + 1 event	Per-scene optimization															
	4DGS	12.76	0.627	194.30	1144.44	37.35	12.25	0.627	196.83	1192.77	36.78	12.01	0.633	320.85	2237.50	36.74
	SFS	15.93	0.695	192.92	1150.29	32.56	15.30	0.695	200.68	1184.41	32.10	15.05	0.655	213.93	1469.48	34.73
	SoM	12.39	0.534	234.73	2190.14	39.85	12.21	0.574	241.41	2633.56	42.04	11.99	0.598	245.80	2669.30	36.74
	D-FUSE (Ours)	20.03	0.234	30.89	342.21	64.77	19.64	0.241	31.72	359.67	64.65	19.28	0.251	34.53	381.91	64.57
Kubric4D-100 1 RGB + 1 event	Feed-forward NVS															
	FILM + GCD	14.11	0.496	82.33	1481.29	50.48	13.74	0.555	91.42	1503.50	48.56	13.55	0.595	91.81	1567.03	45.57
	FILM + MoViES	9.67	0.713	139.43	1367.71	57.12	9.50	0.714	140.13	1378.98	57.23	9.42	0.716	61.36	983.35	57.10
	FILM + TC	12.94	0.401	50.63	953.55	68.05	12.04	0.466	60.40	975.05	68.12	11.94	0.567	61.36	983.35	67.96
	D-FUSE (Ours)	15.60	0.378	19.81	89.99	66.30	15.47	0.385	20.19	96.50	66.17	15.57	0.389	19.28	292.40	66.06

5 Evaluation

Datasets and Metrics. We evaluate D-FUSE on both synthetic and real-world data. For synthetic, we use the Kubric4D [Greff et al. 2022; Van Hoorick et al. 2024] test set (100 scenes, 16 views, 33 frames) and simulate events from the RGB videos with V2E [Hu et al. 2021]. Events are kept at their native rate while RGB is subsampled to 1/2, 1/4, and 1/8 of that rate. The output video is sampled at the full event rate, so the three settings correspond to 2×, 4×, and 8× RGB upsampling. Since per-scene optimization baselines require multiple RGB views and are expensive to train, we evaluate them on a 16-scene subset (Kubric4D-16) with 3 RGB views and 1 event view, which creates challenging sparse viewpoint coverage; feed-forward diffusion baselines are evaluated on the full 100-scene set (Kubric4D-100) with 1 RGB and 1 event view. We report D-FUSE under both setups. Since no existing real-world benchmark provides multi-view RGB+event captures of dynamic scenes, we collect our own real-world data: three dynamic scenes recorded with an RGB and event camera rig (RGB at 30 Hz, events at the native sensor rate), evaluated at an 8× upsample rate. We report PSNR and LPIPS [Zhang et al. 2018] for visual fidelity, FID [Heusel et al. 2017] and FVD [Unterthiner et al. 2019] for perceptual quality at the frame and video levels, and MUSIQ [Ke et al. 2021] for reference-free perceptual quality.

Baselines. We compare D-FUSE against two categories of novel view synthesis methods. (1) *Per-scene optimization*: 4D Gaussian Splatting [Wu et al. 2024] (RGB only), SensorFusionSplatting (SFS) [Zou et al. 2025] (RGB + event), and Shape of Motion (SoM) [Wang et al. 2025] (RGB + depth + scene flow). (2) *Feed-forward NVS*: the

regression-based MoVieS [Lin et al. 2026] (RGB only), and two generative methods, Generative Camera Dolly (GCD) [Van Hoorick et al. 2024] (RGB only), and TrajectoryCrafter (TC) [Yu et al. 2025] (RGB only) as the state-of-the-art. Since these feed-forward baselines assume uniformly sampled RGB inputs, we first upsample the subsampled RGB stream to the event rate using FILM [Reda et al. 2022], a feed-forward RGB frame interpolator, and feed the result into each method. This two-stage pipeline is the closest available proxy for asynchronous multi-sensor NVS. Details of the baseline implementations and evaluation are in the supplementary material.

5.1 Results and Discussion

Synthetic Data Results. Quantitatively, Table 1 reports Kubric4D metrics, with the top block evaluating per-scene optimization on Kubric4D-16 (3 RGB + 1 event) and the bottom evaluating feed-forward methods on Kubric4D-100 (1 RGB + 1 event). D-FUSE outperforms every baseline on PSNR, LPIPS, FID, and FVD across all three asynchrony levels (RGB at 1/2, 1/4, 1/8 of the event rate). Firstly, on Kubric4D-16, D-FUSE improves PSNR by ~4 dB over the strongest per-scene baseline (SFS), while SFS and SoM both fail to fuse the multimodal inputs under such sparse views. Secondly, our metrics are largely stable as the RGB stream is subsampled, with PSNR dropping only ~0.75 dB on Kubric4D-16 and staying essentially flat on Kubric4D-100, confirming that the native-rate event stream compensates for missing RGB frames. Thirdly, among feed-forward baselines, FILM+TC scores ~2 points above us on MUSIQ but trails by ~3 dB in PSNR with an FVD of ~950–985 against our

~90–290, producing visually crisp yet geometrically and temporally inconsistent renderings.

Qualitatively, Fig. 8 shows the visual comparison, with visualizations of the camera poses of RGB, event cameras, and novel views. In the top, view 1 evaluates at input RGB cameras, and views 2–4 are all novel. D-FUSE stays sharp and robust to geometric changes, while SFS produces blur and color-bleed artifacts and SoM leaves unseen regions as a white background. In the bottom, view 1 is the event camera, probing color information learned across modalities. D-FUSE remains consistent and photorealistic everywhere, whereas GCD is incoherent across views and frames, the non-generative MoViS leaves unseen regions as large black holes, and TC fills holes with clean visuals yet loses scene dynamics, with objects hovering in the air across frames. We provide additional frame-by-frame comparison of D-FUSE and ground truth in supplementary material.

Real-World Data Results. In Fig. 9, we evaluate on three real-world scenes captured with our RGB-event rig. In the top scene, GCD fails to generalize to the new views and produces blurry renderings. MoViS and TC incorrectly reconstruct the scene geometry, producing large ghosting artifacts across views and frames. In the middle scene, GCD again produces blurry renderings, while MoViS reconstructs the geometry better but suffers from large holes and artifacts. TC produces sharper renderings but creates unrealistic filling content. In the bottom scene, we present a real-world application analyzing the flight behavior of a hummingbird hawk moth, and we compare against TC, the most competitive baseline. The results show that D-FUSE produces more visually coherent appearance and motion, while TC produces artifacts in appearance and inconsistent motion. Overall, all baselines struggle to interpolate the dynamic motion because they cannot exploit the dense temporal information that event cameras provide, while D-FUSE effectively leverages the event stream and fuses it with the RGB stream to produce sharper and more consistent renderings. Overall, our approach demonstrates robust alignment and fusion on both synthetic and real-world data, including challenging scenarios involving fast motion, sparse viewpoints, and noisy in-the-wild captures. We observe that reconstruction quality remains high when the target views lie within or near the region spanned by the training camera views, while it degrades as the target viewpoints move farther outside this region. Additionally, scenes containing repetitive textures under motion (e.g., a rotating checkerboard) can introduce reconstruction errors. Additional visualizations of indoor and outdoor scenes, and discussion on failure cases, are provided in the supplementary material.

6 Ablation Study

We conduct a comprehensive ablation study to analyze the impact of various design choices of ProbRoPE on the performance of asynchronous input alignment. We focus on the following variants: (i) **RoPE**; (ii) **RoPE with physical timestamps**; (iii) **Uniform Probability**; (iv) **Gaussian Probability**; and (v) **Learned Probability**.

We first conduct a 1D synthetic verification of the five temporal encodings using a random mixture of sinusoids as the ground-truth data. The measurement data is simulated by downsampling the ground-truth and adding random Gaussian noise. We then use a DiT

Table 2. *MSE Error in Interpolation of 1D synthetic data.* We present the MSE error of the interpolation of 1D synthetic data with different temporal encodings. k represents the subsampling factor. Best per column in bold. The results show that ProbRoPE with learned probability modulation achieves the best performance across all subsampling factors, demonstrating its superior ability to handle aliasing effects and reconstruct high-frequency signals.

Encoding	$k=2$	$k=4$	$k=8$
RoPE	0.140	0.451	0.862
RoPE with timestamps	0.004	0.015	0.443
ProbRoPE (Uniform)	0.003	0.025	0.447
ProbRoPE (Gaussian)	0.004	0.035	0.446
ProbRoPE (learned)	0.003	0.009	0.439

conditioned on the measurement data to recover the underlying ground-truth, and report the reconstruction errors in Table 2. The learned ProbRoPE outperforms all other variants in interpolation, demonstrating its superior ability to reconstruct high-frequency signals. Detailed implementation and analysis of the 1D synthetic verification are provided in the supplementary material.

We evaluate the performance of each variant in our multimodal multi-view setup. We train each variant on the data of Bai et al. [2024] for 1.6k iterations and evaluate its performance every 320 iterations. The results are summarized in Fig. 6. *Gaussian* and *Uniform* perform on par with or even worse than standard RoPE. This indicates that fixed, predefined distributions fail to characterize the correct temporal distribution. In contrast, *Learned* modulation progressively pulls ahead and achieves the best final performance. We attribute this to its ability to learn the temporal distribution from data, rather than committing to a hand-designed distribution, which is critical for handling the heterogeneous temporal distribution induced by different sensors.

While ProbRoPE outperforms others on frame-averaged visual quality metrics, these scores do not fully capture how ProbRoPE addresses the temporal asynchrony inherent in dynamic motion. To probe this directly, we compare ProbRoPE against standard RoPE on a motion tracking task on a held-out scene from Bai et al. [2024]. As illustrated in Fig. 7, the scene features fast arm-waving motion and distinct RGB, event, and novel viewpoints, where RGB has a 1/8 temporal resolution compared to the event stream. We generate novel-view videos with 8× temporal interpolation from both ProbRoPE and RoPE, and apply TAPIR [Doersch et al. 2023] to track the subject’s right hand across the ground-truth, ProbRoPE, and RoPE videos, shown as red curves in the figure. The results show that RoPE lifts the arm too early and lowers it spuriously, with a looping trajectory that fails to capture the held-up pose. In contrast, ProbRoPE closely follows the ground-truth hand position across all sampled timesteps, with a smooth trajectory that faithfully captures the arm-waving motion. Table 3 reports the RMS error of the tracked hand position against the ground truth, showing that ProbRoPE reduces the absolute position RMS from 60.01 to 8.01 pixels and the motion-relative RMS from 83.07 to 7.78 pixels, confirming that ProbRoPE more faithfully represents the temporal structure of

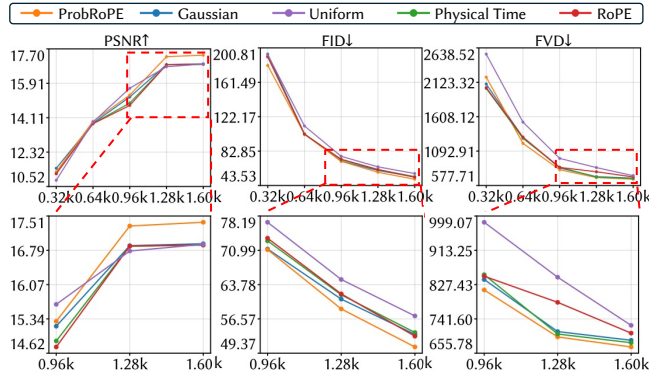


Fig. 6. *Ablation study of ProbRoPE variants on training convergence.* We compare the PSNR/FID/FVD convergence during training of RoPE [Su et al. 2024], RoPE with physical timestamps, and three variants of ProbRoPE (uniform, Gaussian, and learned). We evaluate every 320 iterations over 1.6k training iterations. The learned probability ProbRoPE progressively outperforms the other variants and baselines, demonstrating that learning the temporal distribution of each token effectively models the asynchrony of multimodal inputs.

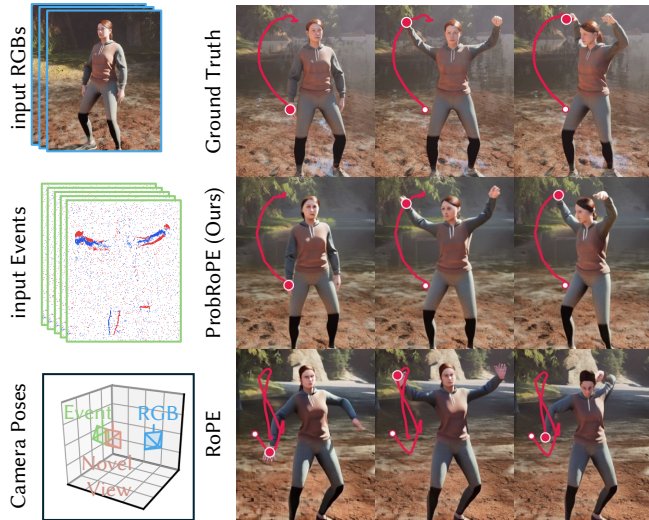


Fig. 7. *Point Tracking Comparison between ProbRoPE and RoPE [Su et al. 2024].* We render a novel view of a person waving her arms with ProbRoPE and RoPE, and run an off-the-shelf 2D point tracker to follow the subject’s right hand. We visualize the input RGB, events, and camera poses on the left, and novel-view frames for ground truth, ProbRoPE, and RoPE on the right. The red trajectory shows the tracked 2D hand positions over time. ProbRoPE successfully synthesizes a temporally coherent video that resembles the ground-truth hand motion, while RoPE leaves the hand too early and lowers the hand spuriously. This demonstrates that ProbRoPE effectively models the temporal asynchrony of multimodal inputs and faithfully captures the temporal structure of dynamic motion.

asynchronous multi-sensor motion, which is critical for accurate reconstruction of fast dynamics.

Table 3. *RMS Errors of Point Tracking of ProbRoPE and RoPE.* We evaluate the RMS error of 2D point tracking of the subject’s right hand in a novel-view video rendered by ProbRoPE and RoPE, compared to the ground-truth video. We report both the absolute RMS error and the motion-relative RMS error. The results show that ProbRoPE significantly reduces the tracking error compared to RoPE, demonstrating that ProbRoPE more faithfully captures the temporal structure of asynchronous multi-sensor motion.

Method	Abs. RMS↓ (px)	Motion-rel. RMS↓ (px)
RoPE	60.01	83.07
ProbRoPE (Ours)	8.01	7.78

7 Conclusion and Future Work

In this work, we presented D-FUSE, a feed-forward video diffusion transformer for asynchronous multimodal sensor fusion and novel view synthesis of high-speed scenes. Our key insight is that conventional positional encodings introduce temporal aliasing under asynchronous sensing, where measurements acquired over different temporal intervals are forced into identical deterministic representations. To address this, we introduced ProbRoPE, a probabilistic extension of rotary position embedding (RoPE) that represents each observation as a distribution over its capture interval. This enables effective alignment across viewpoints and sampling rates, allowing D-FUSE to operate directly on asynchronous heterogeneous measurements and synthesize temporally consistent novel views without per-scene optimization. Beyond novel view synthesis, our formulation opens a broader direction toward scalable and flexible multimodal capture systems that do not rely on tightly synchronized sensor arrays.

We demonstrated the effectiveness of our approach on both synthetic and real-world data, including challenging scenarios involving fast motion, sparse viewpoints, and noisy in-the-wild captures. While stronger geometric consistency, precise camera control, and real-time rendering remain important directions for future work, recent progress in large-scale 4D reconstruction priors [Xie et al. 2026], diffusion camera control [Li et al. 2025b; Yin et al. 2026], and real-time diffusion models [Huang et al. 2025; Millon 2025] suggests promising opportunities for extending our framework. We believe asynchronous multimodal fusion will become increasingly important for applications in robotics, AR/VR, world modeling, and scientific imaging. We hope that probabilistic temporal representations such as ProbRoPE will motivate a broader shift from treating sensor observations as deterministic timestamps to modeling them as distributions over time.

Acknowledgments

We thank Ke Liu and Xiaofan Chen for serving as dancers, and Jim Mahaney and Sanskriti Negi for coordinating the captures. This research is supported by a DARPA Disruptive Idea Award seed grant on neuro-symbolic systems.

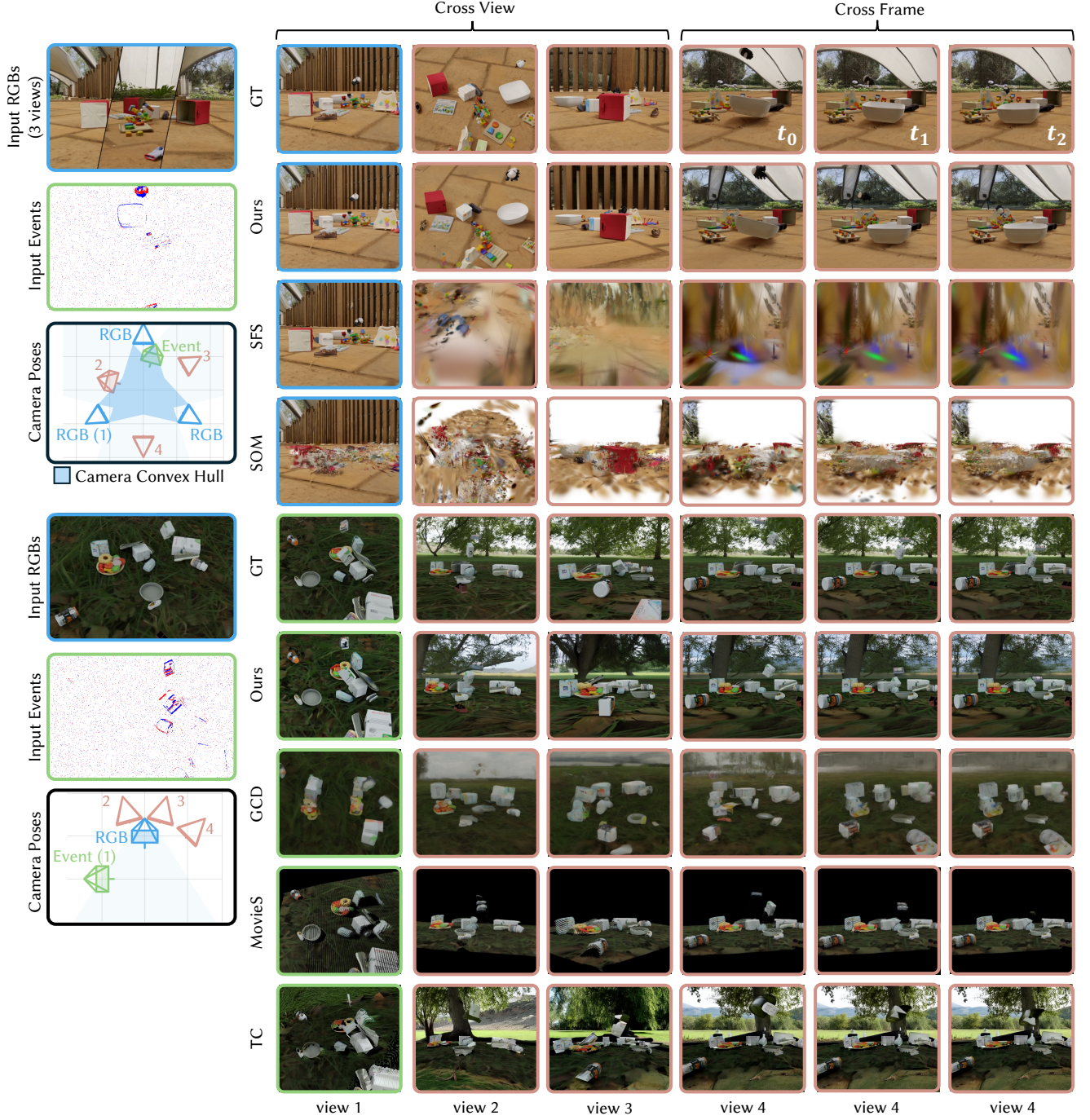


Fig. 8. *Comparison on Synthetic Data.* We compare the visual quality of the synthesized novel views on the synthetic dataset. The first row shows the input RGB, events, and camera poses. The 2nd row shows the synthesized frames at the input RGB views for the top scene, and at the input event views for the bottom scene. The 3rd to 4th rows show the synthesized frames across different novel views. The 5th to 7th rows show the synthesized novel-view frames across different times. The baselines struggle to synthesize high-quality novel views. For per-scene optimization methods like SFS [Zou et al. 2025] and SoM [Wang et al. 2025], they maintain reasonable quality at the input RGB views, but fail to synthesize high-quality novel views. For the feed-forward methods like MoViS [Lin et al. 2026] and TC [Yu et al. 2025], they either leave large occluded areas blank or produce unrealistic content filling. In contrast, D-FUSE outperforms the baselines in visual quality and temporal consistency across views and time, demonstrating its effectiveness in synthesizing high-quality novel views from RGB and event inputs.

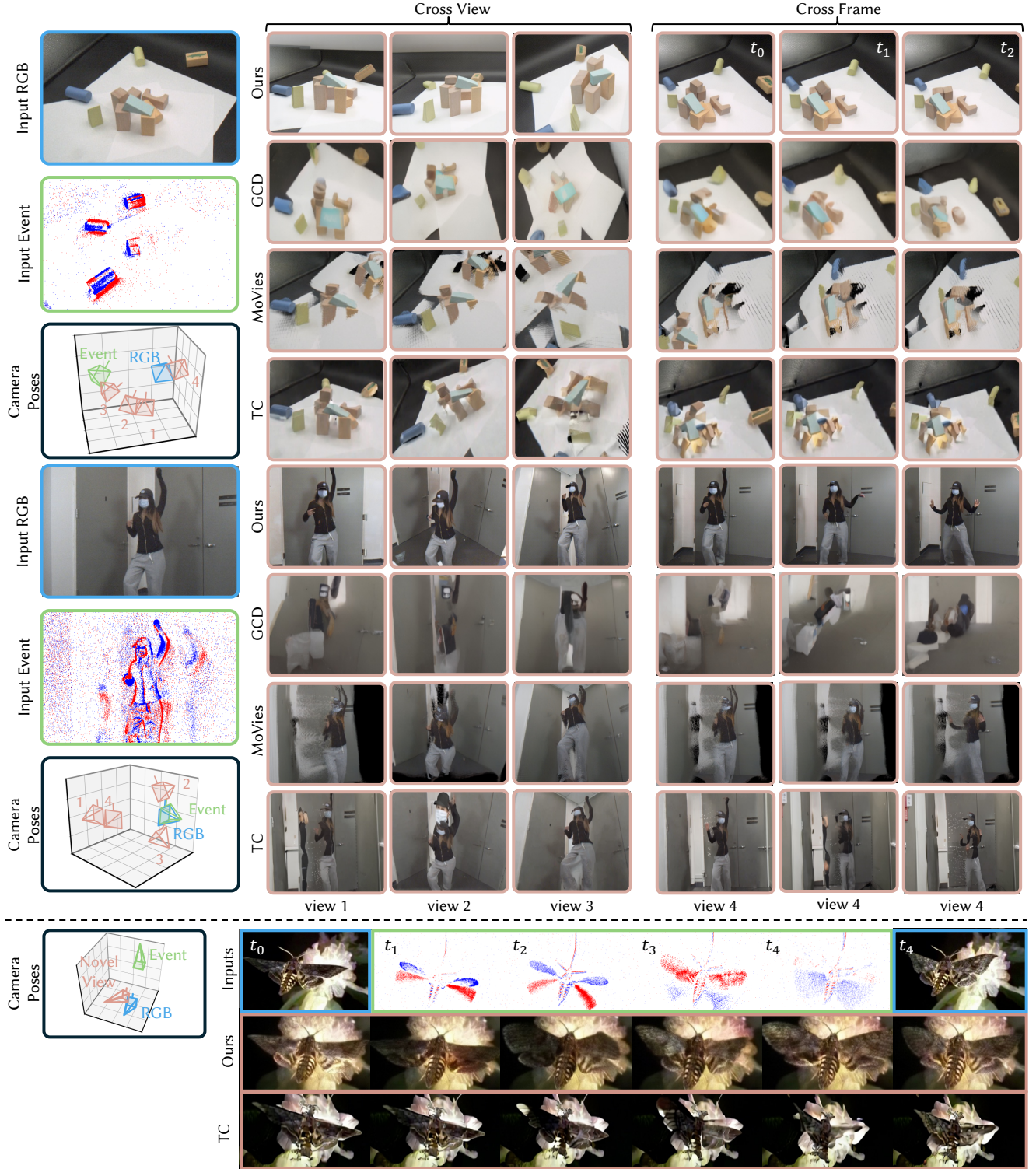


Fig. 9. *Comparison on Real-world Data.* We compare the visual quality of the synthesized novel views on the real-world dataset. The baselines struggle to synthesize temporally and geometrically consistent novel views. In contrast, D-FUSE generates high-quality novel views with better temporal and geometric consistency across views and time, demonstrating its effectiveness in synthesizing high-quality novel views from RGB and event inputs.

References

- Benjamin Attal, Eliot Laidlaw, Aaron Gokaslan, Changil Kim, Christian Richardt, James Tompkin, and Matthew O'Toole. 2021. Törf: Time-of-flight radiance fields for dynamic scene view synthesis. *Advances in neural information processing systems* 34 (2021), 26289–26301.
- Jeongmin Bae, Seoha Kim, Youngsik Yun, Hahyun Lee, Gun Bang, and Youngjung Uh. 2024. Per-gaussian embedding-based deformation for deformable 3d gaussian splatting. In *European Conference on Computer Vision*. Springer, 321–335.
- Jianhong Bai, Menghan Xia, Xiao Fu, Xintao Wang, Lianrui Mu, Jinwen Cao, Zuoqiu Liu, Haoji Hu, Xiang Bai, Pengfei Wan, et al. 2025. Recammaster: Camera-controlled generative rendering from a single video. *arXiv preprint arXiv:2503.11647* (2025).
- Jianhong Bai, Menghan Xia, Xintao Wang, Ziyang Yuan, Xiao Fu, Zuoqiu Liu, Haoji Hu, Pengfei Wan, and Di Zhang. 2024. Syncammaster: Synchronizing multi-camera video generation from diverse viewpoints. *arXiv preprint arXiv:2412.07760* (2024).
- Jonathan T Barron, Ben Mildenhall, Matthew Tancik, Peter Hedman, Ricardo Martin-Brualla, and Pratul P Srinivasan. 2021. Mip-nerf: A multiscale representation for anti-aliasing neural radiance fields. In *Proceedings of the IEEE/CVF international conference on computer vision*. 5855–5864.
- Michael Broxton, John Flynn, Ryan Overbeck, Daniel Erickson, Peter Hedman, Matthew Duvall, Jason Dourgarian, Jay Busch, Matt Whalen, and Paul Debevec. 2020. Immersive light field video with a layered mesh representation. *ACM Transactions on Graphics (TOG)* 39, 4 (2020), 86–1.
- Marco Cannici and Davide Scaramuzza. 2024. Mitigating motion blur in neural radiance fields with events and frames. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 9286–9296.
- Ang Cao and Justin Johnson. 2023. Hexplane: A fast representation for dynamic scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 130–141.
- David Charatan, Sizhe Lester Li, Andrea Tagliasacchi, and Vincent Sitzmann. 2024. pixelsplat: 3d gaussian splats from image pairs for scalable generalizable 3d reconstruction. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 19457–19467.
- Yuedong Chen, Haoqi Xu, Chuanxia Zheng, Bohan Zhuang, Marc Pollefeys, Andreas Geiger, Tat-Jen Cham, and Jianfei Cai. 2024. Mvsplat: Efficient 3d gaussian splatting from sparse multi-view images. In *European conference on computer vision*. Springer, 370–386.
- Carl Doersch, Yi Yang, Mel Vecerik, Dilara Gokay, Ankush Gupta, Yusuf Aytar, Joao Carreira, and Andrew Zisserman. 2023. TAPIR: Tracking any point with per-frame initialization and temporal refinement. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 10061–10072.
- Yilun Du, Yanan Zhang, Hong-Xing Yu, Joshua B Tenenbaum, and Jiajun Wu. 2021. Neural radiance flow for 4d view synthesis and video processing. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE Computer Society, 14304–14314.
- Yuanxing Duan, Fangyin Wei, Qiyu Dai, Yuhang He, Wenzheng Chen, and Baoquan Chen. 2024. 4d-rotor gaussian splatting: towards efficient novel view synthesis for dynamic scenes. In *ACM SIGGRAPH 2024 Conference Papers*. 1–11.
- Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, Axel Sauer, Frederik Boesel, et al. 2024. Scaling rectified flow transformers for high-resolution image synthesis. In *Forty-first international conference on machine learning*.
- Jiemin Fang, Taoran Yi, Xinggang Wang, Lingxi Xie, Xiaopeng Zhang, Wenyu Liu, Matthias Nießner, and Qi Tian. 2022. Fast dynamic radiance fields with time-aware neural voxels. In *SIGGRAPH Asia 2022 Conference Papers*. 1–9.
- Yanhong Fei, Yingjie Liu, Xian Wei, and Mingsong Chen. 2022. O-vit: Orthogonal vision transformer. *arXiv preprint arXiv:2201.12133* (2022).
- Chaoran Feng et al. 2025. E-4DGS: High-Fidelity Dynamic Reconstruction from Multi-View Event Cameras. In *Proceedings of the 33rd ACM International Conference on Multimedia*. Association for Computing Machinery, New York, NY, USA, 10 pages.
- Sara Fridovich-Keil, Giacomo Meanti, Frederik Rahbæk Warburg, Benjamin Recht, and Angjoo Kanazawa. 2023. K-planes: Explicit radiance fields in space, time, and appearance. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 12479–12488.
- Guillermo Gallego, Tobi Delbrück, Garrick Orchard, Chiara Bartolozzi, Brian Taba, Andrea Censi, Stefan Leutenegger, Andrew J Davison, Jörg Conradt, Kostas Daniilidis, et al. 2020. Event-based vision: A survey. *IEEE transactions on pattern analysis and machine intelligence* 44, 1 (2020), 154–180.
- Chen Gao, Ayush Saraf, Johannes Kopf, and Jia-Bin Huang. 2021. Dynamic view synthesis from dynamic monocular video. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 5712–5721.
- Ruiqi Gao, Aleksander Holynski, Philipp Henzler, Arthur Brussee, Ricardo Martin-Brualla, Pratul Srinivasan, Jonathan T Barron, and Ben Poole. 2024. Cat3d: Create anything in 3d with multi-view diffusion models. *arXiv preprint arXiv:2405.10314* (2024).
- Klaus Greff, Francois Belletti, Lucas Beyer, Carl Doersch, Yilun Du, Daniel Duckworth, David J Fleet, Dan Gnanaprasam, Florian Golemo, Charles Herrmann, et al. 2022. Kubric: A scalable dataset generator. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 3749–3761.
- Yoav HaCohen, Nisan Chiprut, Benny Brazowski, Daniel Shalem, Dudu Moshe, Eitan Richardson, Eran Levin, Guy Shiran, Nir Zabari, Ori Gordon, et al. 2024. Ltx-video: Realtime video latent diffusion. *arXiv preprint arXiv:2501.00103* (2024).
- Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. 2017. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in neural information processing systems* 30 (2017).
- Yuhuang Hu, Shih-Chii Liu, and Tobi Delbruck. 2021. v2e: From video frames to realistic DVS events. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 1312–1321.
- Xun Huang, Zhengqi Li, Guande He, Mingyuan Zhou, and Eli Shechtman. 2025. Self Forcing: Bridging the Train-Test Gap in Autoregressive Video Diffusion. *arXiv preprint arXiv:2506.08009* (2025).
- Yi-Hua Huang, Yang-Tian Sun, Ziyi Yang, Xiaoyang Lyu, Yan-Pei Cao, and Xiaojuan Qi. 2024. Sc-gs: Sparse-controlled gaussian splatting for editable dynamic scenes. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 4220–4230.
- Lihan Jiang, Yucheng Mao, Lanning Xu, Tao Lu, Kerui Ren, Yichen Jin, Xudong Xu, Mulin Yu, Jiangmiao Pang, Feng Zhao, et al. 2025. Anysplat: Feed-forward 3d gaussian splatting from unconstrained views. *ACM Transactions on Graphics (TOG)* 44, 6 (2025), 1–16.
- Haian Jin, Hanwen Jiang, Hao Tan, Kai Zhang, Sai Bi, Tianyuan Zhang, Fujun Luan, Noah Snavely, and Zexiang Xu. 2024. Lvsm: A large view synthesis model with minimal 3d inductive bias. *arXiv preprint arXiv:2410.17242* (2024).
- Hanbyul Joo, Hao Liu, Lei Tan, Lin Gui, Bart Nabbe, Iain Matthews, Takeo Kanade, Shohei Nobuhara, and Yaser Sheikh. 2015. Panoptic studio: A massively multiview system for social motion capture. In *Proceedings of the IEEE international conference on computer vision*. 3334–3342.
- Junjie Ke, Qifei Wang, Yilin Wang, Peyman Milanfar, and Feng Yang. 2021. Musiq: Multi-scale image quality transformer. In *Proceedings of the IEEE/CVF international conference on computer vision*. 5148–5157.
- Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 2023. 3D Gaussian splatting for real-time radiance field rendering. *ACM Trans. Graph.* 42, 4 (2023), 139–1.
- Diederik P Kingma and Max Welling. 2014. Auto-encoding variational bayes. In *International Conference on Learning Representations*.
- Simon Klenk, Lukas Koestler, Davide Scaramuzza, and Daniel Cremers. 2023. E-nerf: Neural radiance fields from a moving event camera. *IEEE Robotics and Automation Letters* 8, 3 (2023), 1587–1594.
- Deqi Li, Shi-Sheng Huang, Zhiyuan Lu, Xinran Duan, and Hua Huang. 2024. St-4dgs: Spatial-temporally consistent 4d gaussian splatting for efficient dynamic scene rendering. In *ACM SIGGRAPH 2024 Conference Papers*. 1–11.
- Jiaye Li, Baoyou Chen, Hui Li, Zilong Dong, Jingdong Wang, and Siyu Zhu. 2025a. Head-wise Adaptive Rotary Positional Encoding for Fine-Grained Image Generation. *arXiv preprint arXiv:2510.10489* (2025).
- Ruilong Li, Brent Yi, Junchen Liu, Hang Gao, Yi Ma, and Angjoo Kanazawa. 2025b. Cameras as Relative Positional Encoding. *Advances in Neural Information Processing Systems* (2025).
- Tianye Li, Mira Slavcheva, Michael Zollhoefer, Simon Green, Christoph Lassner, Changil Kim, Tanner Schmidt, Steven Lovegrove, Michael Goesele, Richard Newcombe, et al. 2022. Neural 3d video synthesis from multi-view video. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 5521–5531.
- Zhengqi Li, Simon Niklaus, Noah Snavely, and Oliver Wang. 2021. Neural scene flow fields for space-time view synthesis of dynamic scenes. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 6498–6508.
- Hanxue Liang, Jiawei Ren, Ashkan Mirzaei, Antonio Torralba, Ziwei Liu, Igor Gilitschenski, Sanja Fidler, Cengiz Oztireli, Huan Ling, Zan Gojcic, et al. 2024. Feed-forward bullet-time reconstruction of dynamic scenes from monocular videos. *arXiv preprint arXiv:2412.03526* (2024).
- Chenguo Lin, Yuchen Lin, Panwang Pan, Yifan Yu, Tao Hu, Honglei Yan, Katerina Fragkiadaki, and Yadong Mu. 2026. MoVie: Motion-Aware 4D Dynamic View Synthesis in One Second. In *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*.
- Youtian Lin, Zuoqiu Dai, Siyu Zhu, and Yao Yao. 2024. Gaussian-flow: 4d reconstruction with dynamic 3d gaussian particle. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 21136–21145.
- Yaron Lipman, Ricky TQ Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. 2022. Flow matching for generative modeling. *arXiv preprint arXiv:2210.02747* (2022).
- Jia-Wei Liu, Yan-Pei Cao, Weijia Mao, Wenqiao Zhang, David Junhao Zhang, Jussi Keppo, Ying Shan, Xiaohu Qie, and Mike Zheng Shou. 2022a. Devrf: Fast deformable voxel radiance fields for dynamic scenes. *Advances in Neural Information Processing Systems* 35 (2022), 36762–36775.
- Xingchao Liu, Chengyue Gong, and Qiang Liu. 2022b. Flow straight and fast: Learning to generate and transfer data with rectified flow. *arXiv preprint arXiv:2209.03003* (2022).

- Stephen Lombardi, Tomas Simon, Jason Saragih, Gabriel Schwartz, Andreas Lehrmann, and Yaser Sheikh. 2019. Neural volumes: Learning dynamic renderable volumes from images. *arXiv preprint arXiv:1906.07751* (2019).
- Ilya Loshchilov and Frank Hutter. 2017. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101* (2017).
- Yuanxun Lu, Jingyang Zhang, Tian Fang, Jean-Daniel Nahmias, Yanghai Tsin, Long Quan, Xun Cao, Yao Yao, and Shiwei Li. 2025. Matrix3d: Large photogrammetry model all-in-one. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. IEEE, Piscataway, NJ, USA, 11250–11263.
- Jonathon Luiten, Georgios Kopanas, Bastian Leibe, and Deva Ramanan. 2024. Dynamic 3d gaussians: Tracking by persistent dynamic view synthesis. In *2024 International Conference on 3D Vision (3DV)*. IEEE, 800–809.
- Qi Ma, Danda Pani Paudel, Ajad Chhatkuli, and Luc Van Gool. 2023. Deformable neural radiance fields using rgb and event cameras. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 3590–3600.
- Anagh Malik, Parsa Mirdehghan, Sotiris Noutsias, Kyros Kutulakos, and David Lindell. 2023. Transient neural radiance fields for lidar view synthesis and 3D reconstruction. *Advances in neural information processing systems* 36 (2023), 71569–71581.
- Xingyu Miao, Yang Bai, Haoran Duan, Fan Wan, Yawen Huang, Yang Long, and Yefeng Zheng. 2024. Ctnerf: Cross-time transformer for dynamic neural radiance field from monocular video. *Pattern Recognition* 156 (2024), 110729.
- Ben Mildenhall, Pratul P Srinivasan, Matthew Tancik, Jonathan T Barron, Ravi Ramamoorthi, and Ren Ng. 2021. Nerf: Representing scenes as neural radiance fields for view synthesis. *Commun. ACM* 65, 1 (2021), 99–106.
- Erwan Millon. 2025. *Krea Realtime 14B: Real-time Video Generation*. <https://github.com/krea-ai/realtime-video>
- Fangzhou Mu, Carter Siferman, Sacha Jungerman, Yiquan Li, Mark Han, Michael Gleicher, Mohit Gupta, and Yin Li. 2024. Towards 3D vision with low-cost single-photon cameras. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 5302–5311.
- Alan V Oppenheim, Alan S Willsky, and Syed Hamid Nawab. 1997. *Signals & systems*. Pearson Education.
- Byeongjun Park, Byung-Hoon Kim, Hyungjin Chung, and Jong Chul Ye. 2025. ReDirector: Creating Any-Length Video Retakes with Rotary Camera Encoding. *arXiv preprint arXiv:2511.19827* (2025).
- Keunhong Park, Utkarsh Sinha, Jonathan T. Barron, Sofien Bouaziz, Dan B. Goldman, Steven M. Seitz, and Ricardo Martin-Brualla. 2021a. NeRFies: Deformable Neural Radiance Fields. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 5865–5874.
- Keunhong Park, Utkarsh Sinha, Peter Hedman, Jonathan T Barron, Sofien Bouaziz, Dan B Goldman, Ricardo Martin-Brualla, and Steven M Seitz. 2021b. Hypernerf: A higher-dimensional representation for topologically varying neural radiance fields. *arXiv preprint arXiv:2106.13228* (2021).
- William Peebles and Saining Xie. 2023. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*. 4195–4205.
- Albert Pumarola, Enric Corona, Gerard Pons-Moll, and Francesc Moreno-Noguer. 2021. D-NeRF: Neural Radiance Fields for Dynamic Scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 10318–10327.
- LIU Qingming, Yuan Liu, Jiepeng Wang, Xianqiang Lyu, Peng Wang, Wenping Wang, and Junhui Hou. 2025. MoDGS: Dynamic gaussian splatting from casually-captured monocular videos with depth priors. In *The Thirteenth International Conference on Learning Representations*. OpenReview.net, Appleton, WI, USA, 10 pages.
- Qwen Team. 2026. Qwen3.5: Towards Native Multimodal Agents. <https://qwen.ai/blog?id=qwen3.5>
- Fitsum Reda, Janne Kontkanen, Eric Tabellion, Deqing Sun, Caroline Pantofaru, and Brian Curless. 2022. Film: Frame interpolation for large motion. In *European Conference on Computer Vision*. Springer, 250–266.
- Jeremy Reizenstein, Roman Shapovalov, Philipp Henzler, Luca Sbordone, Patrick Labatut, and David Novotny. 2021. Common Objects in 3D: Large-Scale Learning and Evaluation of Real-life 3D Category Reconstruction. In *International Conference on Computer Vision*.
- Jiawei Ren, Kevin Xie, Ashkan Mirzaei, Hanxue Liang, Xiaohui Zeng, Karsten Kreis, Ziwei Liu, Antonio Torralba, Sanja Fidler, Seung W Kim, et al. 2024. L4gm: Large 4d gaussian reconstruction model. *Advances in Neural Information Processing Systems* 37 (2024), 56828–56858.
- Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 10684–10695.
- Viktor Rudnev, Gereon Fox, Mohamed Elgharib, Christian Theobalt, and Vladislav Golyanik. 2025. Dynamic EventNeRF: Reconstructing General Dynamic Scenes from Multi-view RGB and Event Streams. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. IEEE, Piscataway, NJ, USA, 4866–4876.
- Ruizhi Shao, Zerong Zheng, Hanzhang Tu, Boning Liu, Hongwen Zhang, and Yebin Liu. 2023. Tensor4d: Efficient neural 4d decomposition for high-fidelity dynamic reconstruction and rendering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 16632–16642.
- Roman Shapovalov, Yanir Kleiman, Ignacio Rocco, David Novotny, Andrea Vedaldi, Changan Chen, Filippos Kokkinos, Ben Graham, and Natalia Neverova. 2023. Replay: Multi-modal multi-view acted videos for casual holography. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 20338–20348.
- Liangchen Song, Anpei Chen, Zhong Li, Zhang Chen, Lele Chen, Junsong Yuan, Yi Xu, and Andreas Geiger. 2023. Nerfplayer: A streamable dynamic scene representation with decomposed neural radiance fields. *IEEE Transactions on Visualization and Computer Graphics* 29, 5 (2023), 2732–2742.
- Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Bo Wen, and Yunfeng Liu. 2024. RoFormer: Enhanced transformer with Rotary Position Embedding. *Neurocomputing* 568 (2024), 127063. doi:10.1016/j.neucom.2023.127063
- Stanislaw Szymanowicz, Christian Rupprecht, and Andrea Vedaldi. 2024. Splatler image: Ultra-fast single-view 3d reconstruction. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 10208–10217.
- Fengrui Tian, Shaoyi Du, and Yueqi Duan. 2023. Mononerf: Learning a generalizable dynamic radiance field from monocular videos. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. IEEE, Piscataway, NJ, USA, 17903–17913.
- Thomas Unterthiner, Sjoerd Van Steenkiste, Karol Kurach, Raphaël Marinier, Marcin Michalski, and Sylvain Gelly. 2019. FVD: A new metric for video generation. (2019).
- Basile Van Hoorick, Rundi Wu, Ege Ozguroglu, Kyle Sargent, Ruoshi Liu, Pavel Tokmakov, Achal Dave, Changxi Zheng, and Carl Vondrick. 2024. Generative camera dolly: Extreme monocular dynamic novel view synthesis. In *European Conference on Computer Vision*. Springer, 313–331.
- Team Wan, Ang Wang, Baole Ai, Bin Wen, Chaojie Mao, Chen-Wei Xie, Di Chen, Fei Wu Yu, Haiming Zhao, Jianxiao Yang, et al. 2025. Wan: Open and advanced large-scale video generative models. *arXiv preprint arXiv:2503.20314* (2025).
- Qianqian Wang, Zhicheng Wang, Kyle Genova, Pratul P Srinivasan, Howard Zhou, Jonathan T Barron, Ricardo Martin-Brualla, Noah Snavely, and Thomas Funkhouser. 2021. Ibrnet: Learning multi-view image-based rendering. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 4690–4699.
- Qianqian Wang, Vickie Ye, Hang Gao, Weijia Zeng, Jake Austin, Zhengqi Li, and Angjoo Kanazawa. 2025. Shape of motion: 4d reconstruction from a single video. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. IEEE, Piscataway, NJ, USA, 9660–9672.
- Bennett Wilburn, Neel Joshi, Vaibhav Vaish, Eino-Ville Talvala, Emilio Antunez, Adam Barth, Andrew Adams, Mark Horowitz, and Marc Levoy. 2005. High performance imaging using large camera arrays. In *ACM siggraph 2005 papers*. 765–776.
- Guanjun Wu, Taoran Yi, Jiemin Fang, Lingxi Xie, Xiaopeng Zhang, Wei Wei, Wenyu Liu, Qi Tian, and Xingang Wang. 2024. 4d gaussian splatting for real-time dynamic scene rendering. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. IEEE, Piscataway, NJ, USA, 20310–20320.
- Haoyu Wu, Jingyi Xu, Qiaomu Miao, Dimitris Samaras, and Hieu Le. 2025a. One Attention, One Scale: Phase-Aligned Rotary Positional Embeddings for Mixed-Resolution Diffusion Transformer. *arXiv preprint arXiv:2511.19778* (2025).
- Jay Zhangjie Wu, Yuxuan Zhang, Haithem Turki, Xuanchi Ren, Jun Gao, Mike Zheng Shou, Sanja Fidler, Zan Gojcic, and Huan Ling. 2025b. Difi3d+: Improving 3d reconstructions with single-step diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. IEEE, Piscataway, NJ, USA, 26024–26035.
- Wenqi Xian, Jia-Bin Huang, Johannes Kopf, and Changil Kim. 2021. Space-time neural irradiance fields for free-viewpoint video. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 9421–9431.
- Mingyang Xie, Numair Khan, Tianfu Wang, Naina Dhirga, Seonghyeon Nam, Haitao Yang, Zhuo Hui, Christopher Metzler, Andrea Vedaldi, Hamed Pirsiavash, et al. 2026. LaVR: Scene Latent Conditioned Generative Video Trajectory Re-Rendering using Large 4D Reconstruction Models. *arXiv preprint arXiv:2601.14674* (2026).
- Ziyi Yang, Xinyu Gao, Wen Zhou, Shaohui Jiao, Yuqing Zhang, and Xiaogang Jin. 2024. Deformable 3d gaussians for high-fidelity monocular dynamic scene reconstruction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. IEEE, Piscataway, NJ, USA, 20331–20341.
- Zeyu Yang, Hongye Yang, Zijie Pan, and Li Zhang. 2023. Real-time photorealistic dynamic scene representation and rendering with 4d gaussian splatting. *arXiv preprint arXiv:2310.10642* (2023).
- Minghao Yin, Jiahao Lu, Wenbo Hu, Wang Zhao, Ying Shan, and Kai Han. 2026. SCoPE: Sightline-Coordinate Positional Encoding for Video Diffusion Transformers. *arXiv preprint arXiv:2606.27345* (2026).
- Alex Yu, Vickie Ye, Matthew Tancik, and Angjoo Kanazawa. 2021. pixelnerf: Neural radiance fields from one or few images. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 4578–4587.
- Mark Yu, Wenbo Hu, Jinbo Xing, and Ying Shan. 2025. Trajectorycrafter: Redirecting camera trajectory for monocular videos via diffusion models. In *Proceedings of the IEEE/CVF international conference on computer vision*. 100–111.
- Zehao Yu, Anpei Chen, Binbin Huang, Torsten Sattler, and Andreas Geiger. 2024. Mip-Splatting: Alias-free 3D Gaussian Splatting. *Conference on Computer Vision and Pattern Recognition (CVPR)* (2024).

- Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. 2018. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 586–595.
- Xinjie Zhang, Zhening Liu, Yifan Zhang, Xingtong Ge, Dailan He, Tongda Xu, Yan Wang, Zehong Lin, Shuicheng Yan, and Jun Zhang. 2025. Mega: Memory-efficient 4d gaussian splatting for dynamic scenes. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 27828–27838.
- Zihao Zou, Ziyuan Qu, Xi Peng, Vivek Boominathan, Adithya Pediredla, and Praneeth Chakravarthula. 2025. High-Speed Dynamic 3D Imaging with Sensor Fusion Splatting. arXiv preprint arXiv:2502.04630, 10 pages.